GOOD EVENING EVERYONE

We will start @ 09:10 pm

ARRAY

all are of integer type of info

```
int st1_age = 20;
int st2_age = 21;
int st3_age = 22;
int st4_age = 19;
int st5_age = 20;
```

→ Idea is okay, if we have less no. of students.

100 students → 100 variables → managing 100 variables is a difficult task.

↳ array solve this problem under a single variable name.

↳ Also it's easy to manage.

Array

↳ is a data structure which is used to store similar type of infⁿ in a contiⁿ fashion.

way to store data

Multiple types of an array.

↳ 1D (One Dimensional) array. ⟶

↳ 2D (Two Dimensional) array. (matrix / table)

↳ Multi Dimensional array. (

Syntax of 1D array

denotes it's an array.

datatype array-name [ ] = new datatype [size of the array];

Ex:- Create a 1D array of size '5' of integer type

```
int arr[ ] = new int [5];
```

Range of index in an array of size n

[0, n-1]

index →

index	0	1	2	3	4
value	1	45	5	10	60

arr[0] = 1

5 → [0, 4]

Obs:- If we passing any index value other than the range, it will give error.

↳ Error → ArrayIndexOutOfBoundsException.

User Input for an array.

0	1	2	3	4	5	6	7	8	9
10	20	30	40	50	60	70	80	90	100

```java
import java.io.*;
import java.util.*;

public class Solution {

    public static void main(String[] args) {
        Scanner scn = new Scanner(System.in);
        int sizeOfArray = scn.nextInt(); → sizeOfArray = 10

        int arr[] = new int[sizeOfArray]; → creating an array of size → 10

        for(int i=0;i<sizeOfArray;i++) {
            arr[i] = scn.nextInt();
        }
    }
}
```

Test against custom input

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i=0 < 10 (T)   arr[0] = 10

i=1 < 10 (T)   arr[1] = 20

i=2 < 10 (T)   arr[2] = 30

i=3 < 10 (T)   arr[3] = 40

i=4 < 10 (T)   arr[4] = 50

i=5 < 10 (T)   arr[5] = 60

i=6 < 10 (T)   arr[6] = 70

i=7 < 10 (T)   arr[7] = 80

i=8 < 10 (T)   arr[8] = 90

i=9 < 10 (T)   arr[9] = 100

i=10 < 10 (F)

```java
import java.io.*;
import java.util.*;

public class Solution {

    public static void main(String[] args) {
        Scanner scn = new Scanner(System.in);
        int sizeOfArray = scn.nextInt();

        int arr[] = new int[sizeOfArray];

        for(int i=0;i<sizeOfArray;i++) {
            arr[i] = scn.nextInt();
        }

        for(int i=0;i<sizeOfArray;i++) {
            System.out.println("Element at " + i + "th index is : " + arr[i]);
        }
    }
}
```

↳ How to pass array in fn's.

```java

    public static void printArrayElements(int arr[], int sizeOfArray) {
        for(int i=0;i<sizeOfArray;i++) {
            System.out.println("Element at " + i + "th index is : " + arr[i]);
        }
    }

    public static void main(String[] args) {
        Scanner scn = new Scanner(System.in);
        int sizeOfArray = scn.nextInt();

        int arr[] = new int[sizeOfArray];

        for(int i=0;i<sizeOfArray;i++) {
            arr[i] = scn.nextInt();
        }

        printArrayElements(arr,sizeOfArray);

    }
}
```

↳ How to return an array.

```java

    // Returning an array of 1D of integer type
    public static int [] createAnArray(){
        Scanner scn = new Scanner(System.in);
        int sizeOfArray = scn.nextInt();
        int arr[] = new int[sizeOfArray];
        for(int i=0;i<sizeOfArray;i++) {
            arr[i] = scn.nextInt();
        }

        return arr;
    }

```

→ Array takes default value according to the type of datatype.

↳ integer → 0

↳ boolean → false.

↳ string → null

$n=8$

arr →

0	1	2	3	4	5	6	7
25	15	10	-10	100	20	510	40

max → ~~510~~ ~~510~~ ~~510~~ ~~510~~ ~~510~~ ~~510~~ ~~510~~ -510

if present → return idx

otherwise → return -1;

Linear search

d = -510

```
for(int i=0; i<n; i++) {
    if(arr[i]==d)
        return i
}
return -1;
```

→ if(max < arr[i])
   max = arr[i];

if(min > arr[i])
   min = arr[i];

max - min ;